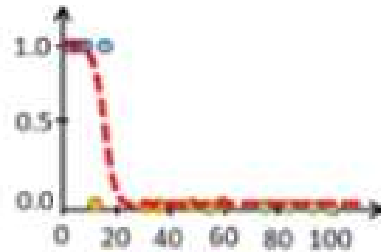


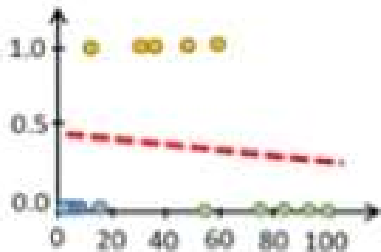
Multiclass Logistic Regression – Softmax function

value	Status
3.2	Low
5.5	Low
7.8	Low
13.4	Low
15.7	Medium
18.9	Medium
20.3	Medium
50.8	Medium
62.4	High
77	High
88.7	High
90.4	High



$$y_1 = w_1x + b_1$$

$$y_1 = -0.4 x X + 6.4$$



$$y_2 = w_2x + b_2$$

$$y_2 = -0.007 x X - 0.3$$

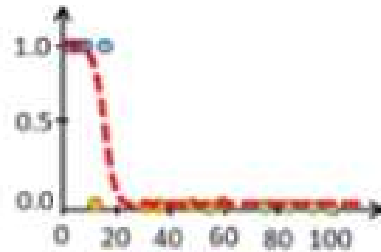


$$y_3 = w_3x + b_3$$

$$y_3 = 0.208 x X - 11.9$$

Multiclass Logistic Regression – Softmax function

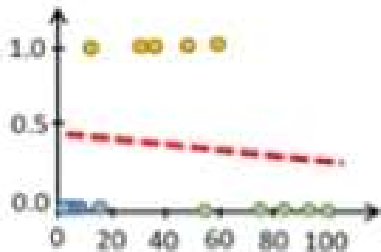
value	Status
3.2	Low
5.5	Low
7.8	Low
13.4	Low
15.7	Medium
18.9	Medium
20.3	Medium
50.8	Medium
62.4	High
77	High
88.7	High
90.4	High



$$y_1 = w_1x + b_1$$

$$y_1 = -0.4x + 6.4$$

$$s(y_1) = \frac{e^{y_1}}{e^{y_1} + e^{y_2} + e^{y_3}}$$



$$y_2 = w_2x + b_2$$

$$y_2 = -0.007x - 0.3$$

$$s(y_2) = \frac{e^{y_2}}{e^{y_1} + e^{y_2} + e^{y_3}}$$



$$y_3 = w_3x + b_3$$

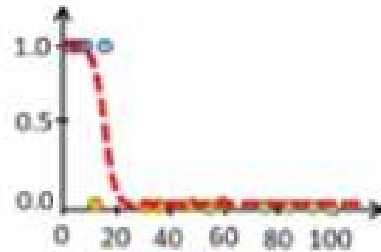
$$y_3 = 0.208x - 11.9$$

$$s(y_3) = \frac{e^{y_3}}{e^{y_1} + e^{y_2} + e^{y_3}}$$

Multiclass Logistic Regression – Softmax function

If value=25

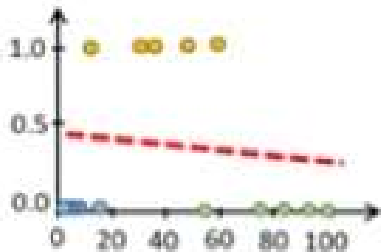
value	Status
3.2	Low
5.5	Low
7.8	Low
13.4	Low
15.7	Medium
18.9	Medium
20.3	Medium
50.8	Medium
62.4	High
77	High
88.7	High
90.4	High



$$y_1 = w_1x + b_1$$

$$y_1 = -0.4x + 6.4 = -3.6$$

$$s(y_1) = \frac{e^{y_1}}{e^{y_1} + e^{y_2} + e^{y_3}} = 0.016$$



$$y_2 = w_2x + b_2$$

$$y_2 = -0.007x - 0.3 = -0.57$$

$$s(y_2) = \frac{e^{y_2}}{e^{y_1} + e^{y_2} + e^{y_3}} = 0.982$$



$$y_3 = w_3x + b_3$$

$$y_3 = 0.208x - 11.9 = -6.73$$

$$s(y_3) = \frac{e^{y_3}}{e^{y_1} + e^{y_2} + e^{y_3}} = 0.002$$